

AI Personal Assistant Evaluation Report

Comparison of an Open-Source Assistant and a Frontier Model Assistant

Prepared for submission to Ollive.ai

Overview

This project compares two AI personal assistants built with the same user experience and conversation flow. The open-source assistant uses Qwen2.5 through Hugging Face Spaces, while the frontier assistant uses Claude Sonnet through the Anthropic API and is deployed with the same Gradio-based interface.

Both assistants support multi-turn conversation, short-term conversational memory using chat history, and basic assistant-like behavior.

Architecture

- **Open-Source Assistant:** Qwen2.5 via Hugging Face Inference API
- **Frontier Assistant:** Claude Sonnet via Anthropic API
- **Interface:** Gradio
- **Memory:** Full conversation history passed on each turn
- **OSS Demo:** <https://huggingface.co/spaces/varshaveeramalli/oss-assisant>
- **Frontier Demo:** <https://huggingface.co/spaces/varshaveeramalli/frontier-assistant>
- **GitHub Repository:** <https://github.com/varshagupta34/ai-assistant-evaluation>

Evaluation Criteria

The assistants were compared across three dimensions:

1. **Hallucination Rate** – factual accuracy on known-answer prompts
2. **Content Safety** – resistance to jailbreaks and harmful requests
3. **Bias & Harmful Outputs** – handling of sensitive or stereotype-based prompts

Results Summary

Category	OSS (Qwen2.5)	Frontier (Claude Sonnet)
Hallucination / Factual Accuracy	8.0/10	9.5/10
Content Safety / Jailbreak Resistance	6.5/10	9.5/10
Bias & Fairness	7.0/10	9.0/10
Overall	7.2/10	9.3/10

Key Findings

- The open-source assistant performed well on basic factual and conversational tasks.
- The frontier assistant was more consistent on difficult prompts and stronger at safe refusal handling.
- The largest performance gap appeared in jailbreak resistance and sensitive-prompt handling.

Recommendation

For production use where reliability and safety are critical, the frontier assistant is the stronger choice. For low-cost experimentation and public demos, the open-source assistant is a practical and effective option.

What I Would Improve

- Add persistent memory using a vector database
- Add tools such as web search and calculator support
- Expand evaluation with more adversarial prompts
- Track latency and cost more formally
- Add observability and safety guardrails